WORKSHOP ON BIOINFORMATICS AND MICROBIOME RESEARCH

From reads to variants

How raw sequencing data becomes a list of mutations, one step and one file format at a time

NM-AIST, Arusha · October 2026

The question: what mutations did this population gain?

Our data come from Richard Lenski's long-term evolution experiment, where *E. coli* populations have been evolving in flasks since 1988. We follow one population, **Ara-3**, sampled at three points in time.

5,000

generations

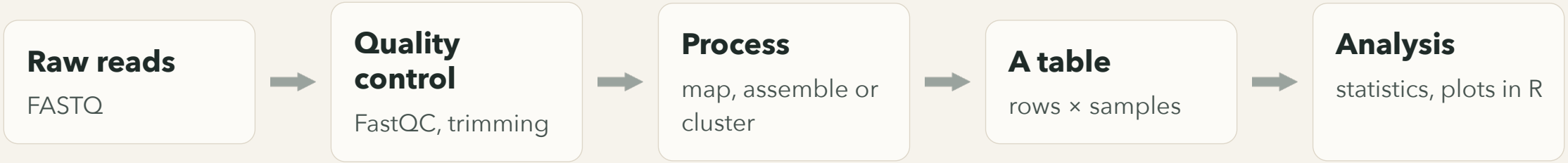
15,000

generations

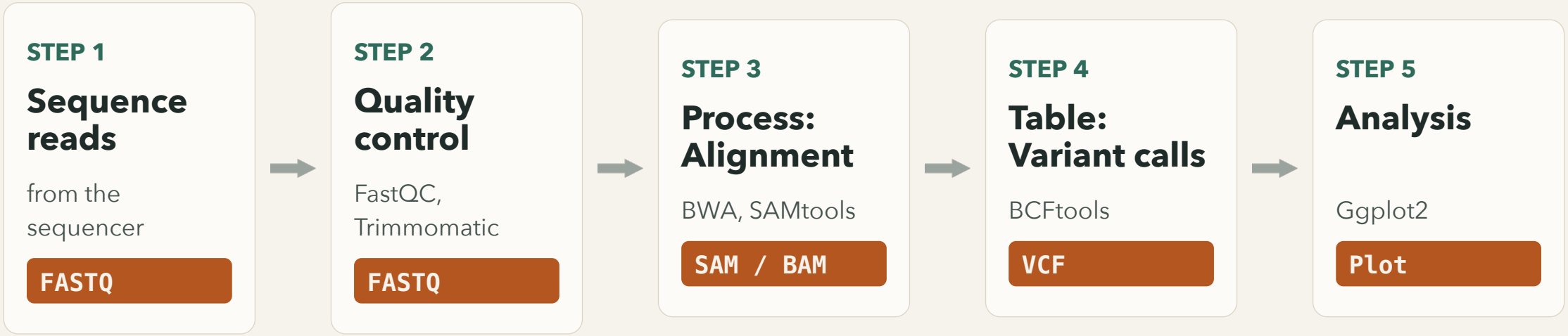
50,000

generations: after the population evolved to use citrate, around 31,000

The same shape fits almost any sequencing project



Five steps, each with its own file format



Each tool reads the file the previous step wrote. If you know the file format, you know where you are in the pipeline.

Step 1: every read is four lines of FASTQ

```
@SRR2584863.1 HWI-ST957:244:H73TDADXX:1:1101:4712:2181/1
TTCACATCCTGACCATTCAGTTGAGCAAAATAGTTCTTCAGTGCCTGTTTAACCGAG…
+
CCCFFFFFGHHHHJIJJJJIJJJIIJJJJIIIJJ … 5C(:A##########
```

Line 1

@ and the read's name

Line 2

The DNA sequence

Line 3

A + separator

Line 4

One quality character per base

Paired-end data come as two files, **_1** and **_2**: the two ends of the same DNA fragment.

Quality scores: each step of 10 is ten times fewer errors

Score	Chance base is wrong	Accuracy	Character
Q10	1 in 10	90%	+
Q20	1 in 100	99%	5
Q30	1 in 1,000	99.9%	?
Q40	1 in 10,000	99.99%	I
Q2	almost a guess	≈ 37%	#

Why characters?

One character per base keeps line 4 the same length as line 2, so every base lines up with its score.

Rule of thumb: **Q30 and above is good**; below Q20 is suspect.

Step 2a: FastQC summarizes millions of reads in one report

```
fastqc *.fastq*
```

Per-base quality

Box plots of quality at each position along the read. Look for where the boxes drop below 20.

Adapter content

Leftover lab sequence from library prep, not your organism. It must be cut off.

Pass, warn, fail

Flags are rough guides. A warning is often normal for your kind of data; read the plot, not just the flag.

Step 2b: trim bad bases and adapters

```
trimmomatic PE SRR2589044_1.fastq.gz SRR2589044_2.fastq.gz \
SRR2589044_1.trim.fastq.gz SRR2589044_1un.trim.fastq.gz \
SRR2589044_2.trim.fastq.gz SRR2589044_2un.trim.fastq.gz \
SLIDINGWINDOW:4:20 MINLEN:25 ILLUMINACLIP:NexteraPE-PE.fa:2:40:15
```

Setting	What it does
SLIDINGWINDOW:4:20	Slide a 4-base window along the read; cut where its average quality drops below Q20
MINLEN:25	Throw away reads shorter than 25 bases after cutting
ILLUMINACLIP:…	Find and remove adapter sequences listed in the adapter file

Two files in, four out: reads whose partner survived (**.trim**), and reads whose partner was thrown away (**un.trim**).

Step 3a: alignment

The reference is the genome of **REL606**, the ancestor strain the experiment started from: about 4.6 million bases.

Each read is a short piece, about 150 bases. Alignment places every piece back on the reference, like matching jigsaw pieces to the picture on the box, while allowing a few differences.

Those differences are what we are looking for.

Once per reference: build an index

```
bwa index ecoli_rel606.fasta
```

Per sample: align both read files

```
bwa mem ecoli_rel606.fasta \
reads_1.fastq reads_2.fastq \
> sample.aligned.sam
```

SAM: one line per read, saying where it landed

Column	Example	Meaning
Read name	SRR2584866.1	Which read this is
Chromosome	CP000819.1	Which reference sequence it hit
Position	1521	Where on the reference it starts
Mapping quality	60	Confidence it belongs there; 0 means it fits several places
CIGAR	150M	How it lines up: 150 matched positions, no gaps

BAM holds exactly the same information, compressed into binary: smaller and faster, but not human-readable.

```
samtools view -S -b in.sam > out.bam
```

Step 3b: sort and index the alignment

Reads come out of the aligner in the order they were sequenced. Variant callers need them in genome order, with an index to jump to any position.

Sort

Reorder reads by position along the genome

```
samtools sort
```

Index

A small .bai file, like a table of contents, so tools can jump to any region

```
samtools index
```

Check

How many reads mapped? A low rate means contamination or the wrong reference

```
samtools flagstat
```

The key idea: stack the reads and look down each column



Every read says G

where the reference has A: a real mutation

One read says C

and the others agree with the reference: probably a sequencing error

Step 4: call variants in three commands

1

```
bcftools mpileup -O b -o raw.bcf -f ecoli_rel606.fasta sorted.bam
```

Count the bases seen at every position: the pileup from the last slide

2

```
bcftools call --ploidy 1 -m -v -o variants.vcf raw.bcf
```

Decide which differences are real. Ploidy 1 because bacteria have one copy of their genome; -v keeps only variant sites

3

```
vcfutils.pl varFilter variants.vcf > final_variants.vcf
```

Filter out weak calls, such as those with too few reads

The result: a VCF, one line per variant

CHROM	POS	REF	ALT	QUAL	DP (depth)
CP000819.1	1,521	C	T	207	9
CP000819.1	1,612	A	G	225	13
CP000819.1	9,092	A	G	225	14
CP000819.1	22,257	C	T	127	5

POS, REF, ALT: where it is, what the ancestor had, what this sample has

QUAL: confidence in the call, on the same Phred scale as read quality

DP: how many reads covered the site. The last row rests on only 5

Rows from sample SRR2584866 in the Data Carpentry lesson; the real file has more columns.

Four ways a variant call can fool you

Too few reads

With 3 reads, one error looks like a third of the evidence. Check depth before trusting a call.

Repeated regions

A read that fits several places gets mapping quality near 0, and its "variants" may be misplaced.

The wrong reference

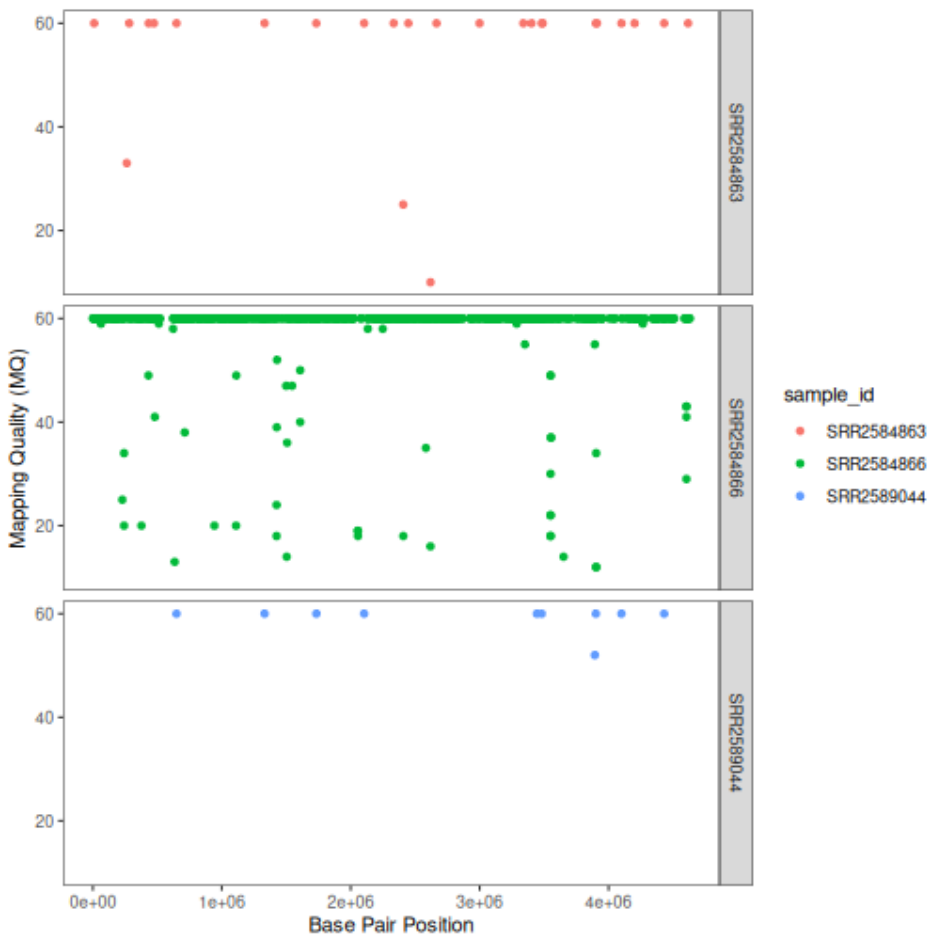
If your sample is too different from the reference, many reads will not map, and the differences you most care about are missed.

Lab artifacts

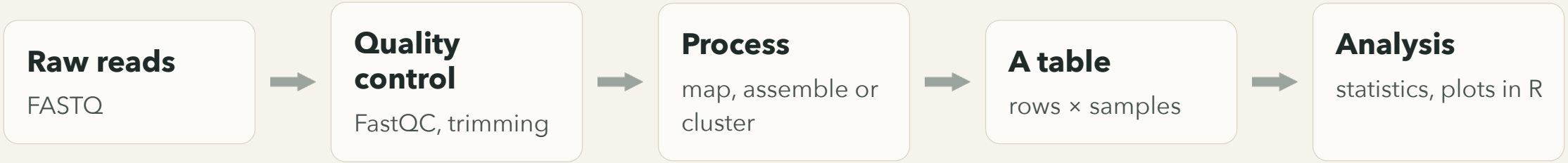
Adapters, PCR copies of the same molecule, and contamination all create false signal. QC exists to catch them.

Step 5: analyze and plot results

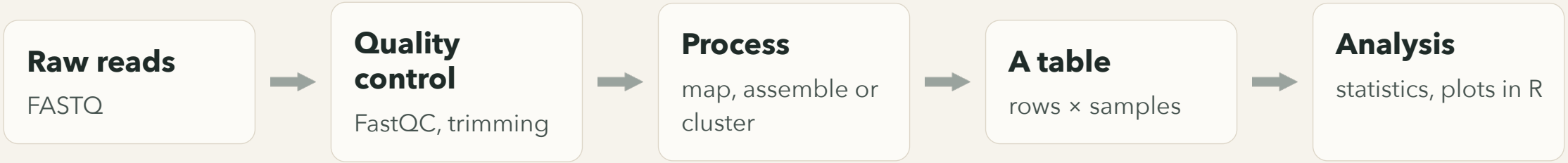
It is straightforward to plot your results in **ggplot2**.



The same shape fits almost any sequencing project



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Data type	Process	The table	Analysis in R
Resequencing (this week)	BWA, then BCFtools	Variants (VCF)	ggplot2
RNA-seq	HISAT2 or STAR; or Salmon	Gene counts	DESeq2
Amplicon (16S, ITS)	DADA2	ASV table	phyloseq
Shotgun metagenomics	Kraken2	Taxon abundances	phyloseq
RAD-seq, GBS	Stacks or ipyrad	Genotypes (VCF)	adegenet, trees

The whole pipeline on one page

Step	Tool	In	Out
Look at quality	FastQC	FASTQ	HTML report
Trim	Trimmomatic	FASTQ	FASTQ (cleaner)
Align	BWA	FASTQ	SAM, then BAM
Sort and index	SAMtools	BAM	sorted BAM + .bai
Call variants	BCFtools	sorted BAM	VCF